

Wireless Networking Assignment

Comprehensive report covering Wi-Fi standards, network architecture, environment scanning, and SSID security.

TASK 1

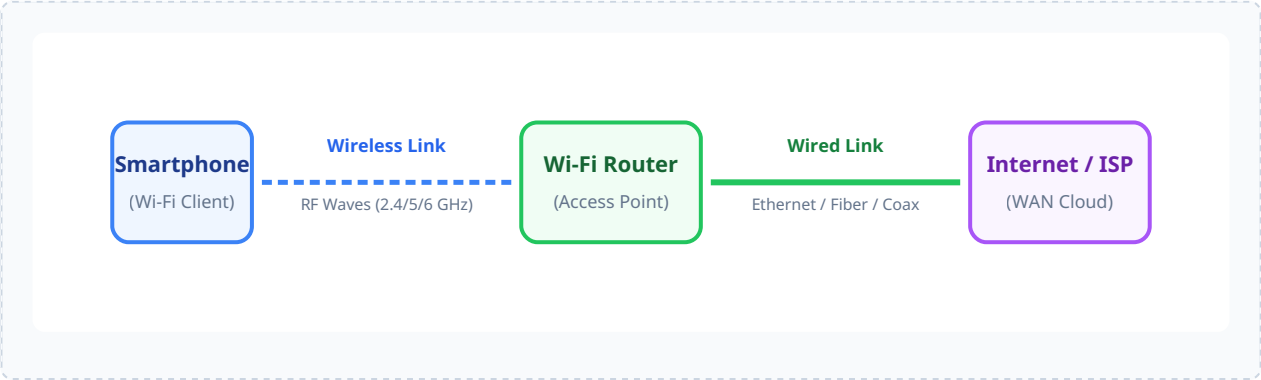
Popular Wi-Fi Standards Comparison

Wi-Fi standards evolve over time to deliver higher throughput, lower latency, and greater connection density in crowded radio frequency environments. Below is a comparison of three prominent standards:

Standard	Frequency Bands	Max Speed	Key Difference & Technological Feature
802.11n (Wi-Fi 4)	2.4 GHz & 5 GHz	Up to 600 Mbps	Introduced MIMO (Multiple-Input Multiple-Output) technology, utilizing multiple antennas simultaneously to multiply data throughput and improve range.
802.11ac (Wi-Fi 5)	5 GHz only	Up to 1.3+ Gbps	Operates exclusively on the cleaner 5 GHz band and introduced wider channel bandwidths (up to 80/160 MHz) along with high-density 256-QAM modulation.
802.11ax (Wi-Fi 6 / 6E)	2.4, 5, & 6 GHz	Up to 9.6 Gbps	Revolutionized multi-device efficiency by introducing OFDMA (Orthogonal Frequency-Division Multiple Access) and Target Wake Time (TWT) for IoT power savings.

TASK 2 Home Network Architecture Diagram

The diagram below illustrates how a smartphone connects wirelessly to a home Wi-Fi router, which subsequently bridges traffic across the wired local and wide-area networks to reach the internet.



Legend: Dashed blue line represents the unguided radio frequency (wireless) medium. Solid green line represents guided physical cabling (wired).

TASK 3 Wi-Fi Network Scan & Selection Analysis

A wireless scan captured the following nearby access points in a typical residential/commercial environment:

Network SSID	Signal Strength (RSSI)	Security Type	Band / Channel
HomeNet_Secure_5G	-45 dBm (Excellent)	WPA3-Personal	5 GHz (Ch 36)
Public_Coffee_Guest	-68 dBm (Fair)	Open (No Security)	2.4 GHz (Ch 6)
Neighbor_WiFi_Ext	-82 dBm (Poor)	WPA2-Personal	2.4 GHz (Ch 11)

Network Selection & Rationale:
I would choose **HomeNet_Secure_5G**. This network offers the strongest signal strength (-45 dBm), ensuring high data rates, minimal packet loss, and robust throughput. Furthermore, it employs **WPA3-Personal**, the most advanced wireless security standard available, which protects against offline dictionary attacks and provides individualized data encryption for each connected device.

TASK 4 Understanding SSID Hiding

What is SSID Hiding?

SSID (Service Set Identifier) hiding is a router configuration option that disables the broadcasting of the network name in periodic *beacon frames*. While normal networks announce their presence automatically, a hidden network requires users or devices to manually input both the exact SSID and security credentials to establish a connection.

Advantage: Reduced Casual Visibility

Hiding your SSID removes your network name from casual Wi-Fi drop-down menus on neighboring devices. This discourages casual bystanders or uninvited guests from attempting to connect or target your router with basic enumeration scripts.

Disadvantage: False Security & Privacy Leak

SSID hiding provides zero cryptographic protection. Your device must constantly broadcast probe requests searching for the hidden SSID ("Is HomeNet nearby?"), allowing attackers to easily sniff the SSID in plaintext and even track your device's physical movements.